4.1.3. GLP-1 Receptor Agonists

GLP-1 receptor agonists (GLP-1RAs) include liraglutide, semaglutide, and dulaglutide. These drugs act like natural GLP-1: they increase insulin release when glucose is high, lower glucagon, slow stomach emptying, and make patients feel full sooner [Israel2026]. RA patients without diabetes also show an impaired incretin-insulin axis, as found in 151 patients compared with 210 sex-matched controls, which ties this hormone system to RA [TejeraSegura2017]. GLP-1 receptors are also present on monocytes, macrophages, and T and B lymphocytes, so these drugs may affect the immune response directly [Israel2026].

In a single-centre retrospective chart review of overweight or obese patients with RA (BMI ≥ 27 kg/m²), 173 patients who took semaglutide or tirzepatide (a dual GIP/GLP-1 receptor agonist) were compared with 42 patients who were prescribed a GLP-1RA but never took it [Kellner2025]. Disease activity was graded from clinic notes on a 0–3 scale (remission to severe), not with a validated score such as DAS28 [Kellner2025]. On this scale, the treated group changed by only −0.03 versus +0.20 in controls ($P = 0.03$), a difference that is statistically significant but probably too small to matter clinically [Kellner2025]. Pain fell by 0.6 cm on a visual analogue scale in the treated group but rose by 1.3 cm in controls ($P < 0.001$) [Kellner2025]. ESR and CRP fell within the treated group, but not more than in controls [Kellner2025]. The drop in pain was not linked to how much weight patients lost, so it may come from effects other than weight loss. The control group was also small, and the reason these patients did not take the drug was unknown [Kellner2025].

Not all data point to a benefit. A large population-based case-control study of 4,535 RA patients and 22,675 controls found that use of semaglutide or liraglutide in the 10 years before diagnosis was linked to higher odds of new-onset RA (adjusted odds ratio [95% CI]: 1.64 [1.26–2.16] and 1.58 [1.22–2.04], respectively), while the result for dulaglutide was not significant [Israel2026]. This link was seen mainly with short use (≤6 months), and the authors suggest the drugs may cause a short-lived rise in inflammation early in treatment [Israel2026]. They also note that reverse causality and confounding by indication may explain part of this result, because obesity, metabolic syndrome and diabetes, the main reasons for giving these drugs, are risk factors for RA on their own [Israel2026].

In short, the evidence on GLP-1RAs in RA is small and comes mostly from observational studies with mixed results. We found no randomized trials of these drugs in RA. For now, GLP-1RAs cannot be recommended for RA, and randomized trials are needed to test them.